

F24-121-D-GrociDish

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

This document outlines "GrociDish," a personalized grocery and meal planning app with nutritional management features. It provides an in-depth overview of the app, including the problem statement, scope, and key stakeholders. The document offers a clear framework for understanding the desired outcomes and overall direction of the project, serving as a comprehensive guide to the project's objectives, scope, and methodologies. This document is designed for a variety of stakeholders. These include developers, project managers, educational administrators, instructors, students, and potential system users. It is designed to provide a thorough grasp of the system's needs and functionality.

1.1 Problem Statement

In Pakistan, maintaining a healthy lifestyle can be challenging due to the lack of apps focusing on local cuisine and dietary habits. Many families struggle to plan grocery lists that are both budget-friendly and nutritionally balanced, leading to unhealthy eating habits or overspending. Home cooks often lack accurate tools to calculate the nutritional content of their recipes, making it difficult to ensure meals meet dietary needs. Additionally, individuals with specific dietary restrictions or preferences face challenges in finding suitable recipes, and food waste is common due to unused ingredients or improper grocery planning. This highlights the need for a solution that not only addresses nutritional management but also integrates local Pakistani cuisine, budget constraints, and personalized grocery generation tailored to families preferences.

1.2 Scope

The app will feature a personalized grocery list generation system that takes into account user preferences, budget, and categories such as dietary needs. For managing recipes, users will have access to a variety of pre-existing options with complete nutritional information, including those specifically tailored for conditions like diabetes, hypertension, and obesity. Calories will be recommended based on each user's Body Mass Index (BMI) and Ideal Body Weight (IBW), ensuring portion control that aligns with individual health goals. Nutritional recommendations, including the balance of macro-nutrients (carbohydrates, fats, and proteins), will be adjusted according to the user's activity levels. This will optimize health outcomes by ensuring that the diet aligns with user's lifestyle and specific needs. The app will recommend recipes based on the ingredients available from the user's grocery list, providing a more customized experience. Additionally, the app will also feature daily caloric tracking for each family member, providing insights into their overall consumption patterns for better health management. However, the app does not calculate nutritional information in real time, meaning users will not receive immediate updates on the nutritional content of their selected ingredients individually. Instead, the recipes will have pre-calculated values. Similarly, the app does not generate new recipes on the fly based on user inputs or available ingredients. Instead, it relies on a pre-existing database of recipes, from which it suggests the most appropriate meal.

1.3 Modules

1.3.1 Personalized Grocery List Generation

This module enables users to generate grocery lists tailored to their preferences (like brands) and budget,

1. Feature 1: Allows users to create grocery lists based on personal preferences .
2. Feature 2: Incorporates budget constraints to generate a cost-effective grocery list that balances nutrition and affordability.
3. Feature 3: Provides brand preferences based on budget, allowing users to select from local and affordable grocery brands for their purchases.

1.3.2 Grocery List Modification via Chatbot

This module integrates a chatbot that assists family members in managing their grocery lists. The chatbot performs various operations such as retrieving, adding, and updating items in the grocery list, ensuring an interactive and convenient experience for users.

1. Feature 1: Allows user to append items to their grocery list via a chat interface. Users can simply input commands like "Add item".
2. Feature 2: Retrieves the current grocery list, providing users with an updated view of the items already added.
3. Feature 3: Incorporates natural language processing to enhance user interaction, making it easier to perform operations mentioned above.

1.3.3 Ingredient-Based Recipe Suggestions

This module recommends recipes based on the ingredients available from the generated grocery list. It enhances customization by offering recipe suggestions that align with what users already have, making meal planning more convenient and efficient.

1. Feature 1: Suggests recipes based on ingredients from the user's grocery list, ensuring minimal food waste and efficient meal planning.
2. Feature 2: Allows users to filter recipe suggestions by dietary preferences or health conditions.

1.3.4 Meal Recommendation System

This module offers users a curated selection of recipes from the *Calorics: Practical Manual of Clinical Nutrition* book, with detailed nutritional information. It also provides recipes suited for health conditions (diabetes, hypertension and obesity).

1. Feature 1: Provides Meals tailored to health conditions like diabetes, hypertension, and obesity, ensuring users can manage their dietary needs effectively.
2. Feature 2: Includes alternative Meal options for variety while maintaining nutritional balance.
3. Feature 3: Nutritional calculations (Total Calories, Fats, carbs, protein and per serving calories) are done with the help of a nutritionist to ensure accuracy and suitability for local cuisine.

4. Feature 4: Users can access a comprehensive database of pre-existing recipes.

1.3.5 Calories Recommendation System

This module calculates recommended serving sizes for each family member, ensuring portion control tailored to individual health goals. It uses the Harris-Benedict equation to personalize calorie and nutrient intake recommendations.

1. Feature 1: Takes user input including age, weight, height, gender and activity level.
2. Feature 2: Calculates BMI (Body Mass Index) and IBW (Ideal Body Weight) for accurate health assessments.
3. Feature 3: Uses the Harris-Benedict equation to calculate daily calorie needs based on the individual's activity level, aiding in meal planning.
4. Feature 4: Recommends for each family member, detailing daily calories, carbohydrates, protein, and fat intake for optimal nutrition.

1.3.6 Caloric Tracking

This module tracks the daily caloric intake of each family member, providing insights into overall consumption and helping users monitor their diet to make healthier choices. It allows for personalized tracking with optional privacy controls.

1. Feature 1: Each family member can have an optional, password-protected calorie tracker, ensuring privacy and individualized monitoring.
2. Feature 2: Tracks daily caloric intake giving users a comprehensive view of their diet.
3. Feature 4: Provides a visual breakdown of calorie sources, making it easier for users to identify unhealthy eating patterns or nutrient deficiencies.

1.4 User Classes and Characteristics

Mentioned below are the identified user classes that will use GrociDish,

User class	Description
Pakistani Families	These are the primary users of GrociDish.They are looking for a convenient way to plan their grocery shopping and meals while staying within a budget. They value personalized grocery lists and Meals that cater to their dietary needs and preferences.
Home Cooks	Individuals who prepare meals at home and need assistance in planning nutritionally balanced meals. They benefit from detailed nutritional information and recipe suggestions based on available ingredients.They can benefit from the wide variety of recipes available
Individuals with Dietary Restrictions	Users who have specific health conditions such as diabetes, hypertension, or obesity. They require specialized recipes and guidance to maintain a healthy lifestyle .They can benefit from the therapeutics recipes available in our app with nutritional information.
Health-Conscious Individuals	Users who are focused on maintaining a healthy lifestyle. They use the app to track their daily caloric intake and ensure their meals are nutritionally balanced.

Table 1.1: User Classes and Characteristics for GrociDish

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of GrociDish.

2.1 Use-case Diagram

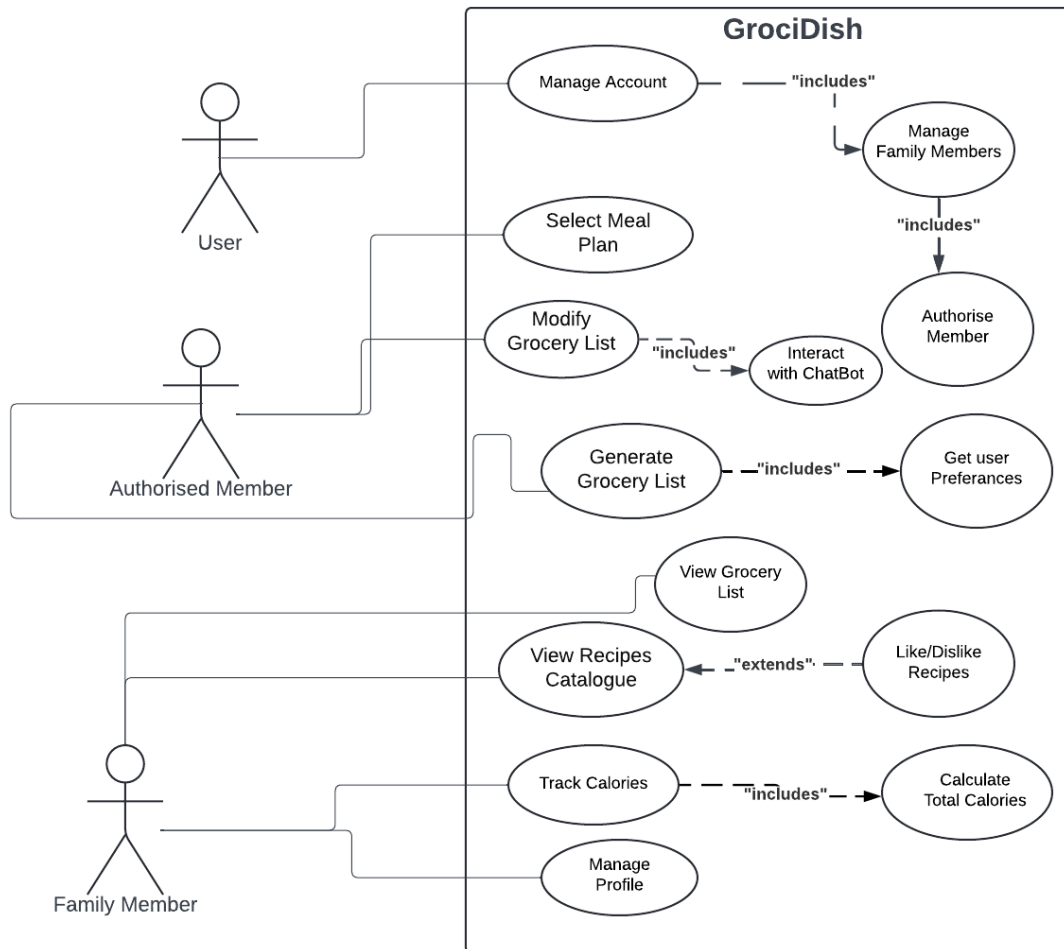


Figure 2.1: GrociDish Use-Case Diagram

2.2 Detailed Use Cases

2.2.1 Manage Account

Use Case Name: Manage Account

Scope: GrociDish System

Level: User Goal

Primary Actor: User(Master User)

Stakeholders and Interests:

- **User:** Wants to create an account and add family members.
- **Family Members:** Want to be added to the system and have appropriate permissions.

Pre-conditions: User has access to the GrociDish application and Internet connection.

Post-conditions: User account is created, and family members are added with appropriate permissions.

Main Success Scenario:

Step	User Action/System Response
1	The user accesses the GrociDish app.
2	The system presents the user with a Sign-up option.
3	The user selects the Sign-Up option.
4	The system displays the User Registration form, prompting the user to enter the necessary information.
5	User enters email, username, and password.
6	The system validates the entered data. Once the user's data passes validation, the system securely stores the user's information in the database.
7	User adds family members.
8	User assigns permissions for grocery list generation and Recipe selection.
9	System confirms account creation and family member addition.
10	The user can now log in using their registered username and password.
11	The user can update and delete account after successful login.

Table 2.1: Main Success Scenario for Manage User Account

Extensions:

- **1a.** If the email is already in use, the system prompts the user to provide a different email or log in with the existing account.
- **6a.** If any field in the registration form is missing or invalid, the system highlights the errors and prompts the user to correct them.
- **8a.** If the user skips assigning permissions to family members, the system prompts for at least one action to be assigned.

2.2.2 Manage Profile

Use Case Name: Manage Profile

Scope: GrociDish System

Level: User Goal

Primary Actor: Family Member

Stakeholders and Interests:

- **Family Member:** Wants to enter personal details for accurate tracking.
- **System:** Needs accurate data for personalized recommendations.

Preconditions: Family member is added to the system by the primary user.

Post-conditions: Family member's personal details are saved in the system.

Main Success Scenario:

Step	User Action/System Response
1	Family member logs into the GrociDish system using the credentials provided by the master user.
2	The system prompts the family member to enter personal details such as name, age, weight, height and activity level.
3	The family member enters their personal information.
4	The system validates the entered data and calculates BMI,IBW and Calculate the members Calories.
5	The system saves the family member's personal details for future use in recommendations.

Table 2.2: Main Success Scenario for Manage Profile

Extensions:

- **1a.** If any detail is missing or invalid, the system prompts the family member to provide the required information.

2.2.3 View Recipe Catalogue

Use Case Name: View Recipe Catalogue

Scope: GrociDish System

Level: User Goal

Primary Actor: Family Member

Stakeholders and Interests:

- **Family Member:** Wants to view recipes and like/dislike recipes.
- **System:** Uses this data for personalized recommendations.

Preconditions: Family Member is logged into the GrociDish system.

Post-conditions: User's recipe preferences are saved and used for personalized recommendations.

Main Success Scenario:

Step	User Action/System Response
1	The Family Member browses the recipe catalog on the app.
2	The Family Member selects a recipe and can like or dislike it.
3	The system saves the recipe preferences for future recommendations.

Table 2.3: Main Success Scenario for View Recipe Catalogue

Extensions:

- **2a.** If the user tries to like or dislike the same recipe multiple times, the system prevents duplicate actions.

2.2.4 Grocery List Generation

Use Case Name: Grocery List Generation

Scope: GrociDish System

Level: User Goal

Primary Actor: Authorized Family Member

Stakeholders and Interests:

- **Authorized Family Member:** Wants to generate a grocery list within a set budget.
- **Family Members:** Want to ensure the list meets their needs.
- **System:** Needs to generate an optimized list based on preferences and budget.

Preconditions: Authorized family member is logged into the GrociDish system and authenticated.

Post-conditions: Grocery list is generated and saved.

Main Success Scenario:

Step	User Action/System Response
1	The authorized family member enters the budget for grocery shopping and set brand preference.
2	The system generates a grocery list based on the budget and family brand preferences.
3	Family members review the grocery list and modify it if necessary.
4	The system saves the final grocery list.

Table 2.4: Main Success Scenario for Grocery List Generation

Extensions:

- **2a.** If the budget is insufficient to meet the needs, the system suggests budget adjustments and prompts.
- **3a.** If Authorized family member modify the list, the system updates the list accordingly.

2.2.5 Track Calories

Use Case Name:Track Calories

Scope: GrociDish System

Level: User Goal

Primary Actor: Family Member

Stakeholders and Interests:

- **Family Member:** Wants to track daily calorie intake accurately.
- **System:** Needs accurate data for health recommendations.

Pre-conditions: Family member is logged into the system.

Post-conditions: Daily calorie intake is tracked and visualized.

Main Success Scenario:

Step	User Action/System Response
1	The family member logs into the GrociDish application.
2	The system displays the dashboard with an option to track calories.
3	The family member selects the calorie tracking option.
4	The system prompts the user to enter individual servings of food consumed.
5	The family member inputs the details of each food item, including serving size and type.
6	The system validates the entered data and calculates the total daily calorie intake.
7	The system displays the calculated calorie intake and provides a visual representation (e.g., bar chart or pie chart) on the dashboard.
8	The family member reviews the visual representation and can see their daily intake compared to recommended levels.
9	The system suggests dietary adjustments if calorie intake exceeds recommended levels.
10	The family member can log out or continue tracking further meals for the day.

Table 2.5: Main Success Scenario for Calorie Tracking

Extensions:

- **1a.** If serving data is incomplete, the system prompts for additional information.
- **2a.** If calorie intake exceeds recommended levels, the system provides alerts.

2.2.6 Select Meal Plan**Use Case Name:** Select Recipe**Scope:** GrociDish System**Level:** User Goal**Primary Actor:** Authorized Family Member**Stakeholders and Interests:**

- **Authorized Family Member:** Wants to select recipes for meal planning.
- **Family Members:** Want to ensure selected recipes meet their dietary preferences.
- **System:** Needs to track selected recipes for calorie tracking.

Pre-conditions: Authorized Family Member is logged into the system and has access to select.

Post-conditions: Selected recipes are saved and used for meal planning.

Main Success Scenario:

Step	User Action/System Response
1	The Authorized Family Member logs into the GrociDish app and clicks on Meal recommendation option.
2	The system displays the recommended Meals.
3	The Authorized Family Member browses through the available Meals for healthy members and for members with health conditions (diabetes,hypertension,obesity).
4	The Authorized Family Member selects a desired Meal.
5	The system saves the selected Meal.
6	The system updates the meal plan and grocery list based on the selected recipe.
7	The system confirms the successful selection and updates to the user.

Table 2.6: Main Success Scenario for Meal Selection

Extensions:

- **2a.** If the recipe is already selected, the system notifies the user.
- **2b.** If the recipe does not meet dietary preferences, the system suggests alternatives.

2.2.7 Modify Grocery List

Use Case Name: Modify Grocery List

Scope: GrociDish System

Level: User Goal **Primary Actor:** Authorized Family Member

Stakeholders and Interests:

- **Authorized Family Member:** Wants to ensure the grocery list meets the family's needs and budget.
- **Family Members:** Want to ensure their preferences are included.
- **System:** Needs to maintain an accurate and optimized grocery list.

Pre-conditions: Authorized family member is logged into the system and has access to the grocery list.

Post-conditions: Grocery list is updated and saved with the modifications.

Main Success Scenario:

Step	User Action/System Response
1	The authorized family member logs into the GrociDish application.
2	The system displays the current grocery list.
3	The authorized family member reviews the grocery list.
4	The authorized family member makes modifications (adds/removes items).
5	The system updates the grocery list with the modifications.
6	The system saves the updated grocery list.
7	The system confirms the successful updates to the user.

Table 2.7: Main Success Scenario for Modifying Grocery List

Extensions:

- **2a.** If an item is not available, the system suggests alternatives.
- **2b.** If the budget is exceeded, the system alerts the user and suggests adjustments.

2.3 System Sequence Diagrams

2.3.0.1 Manage Account

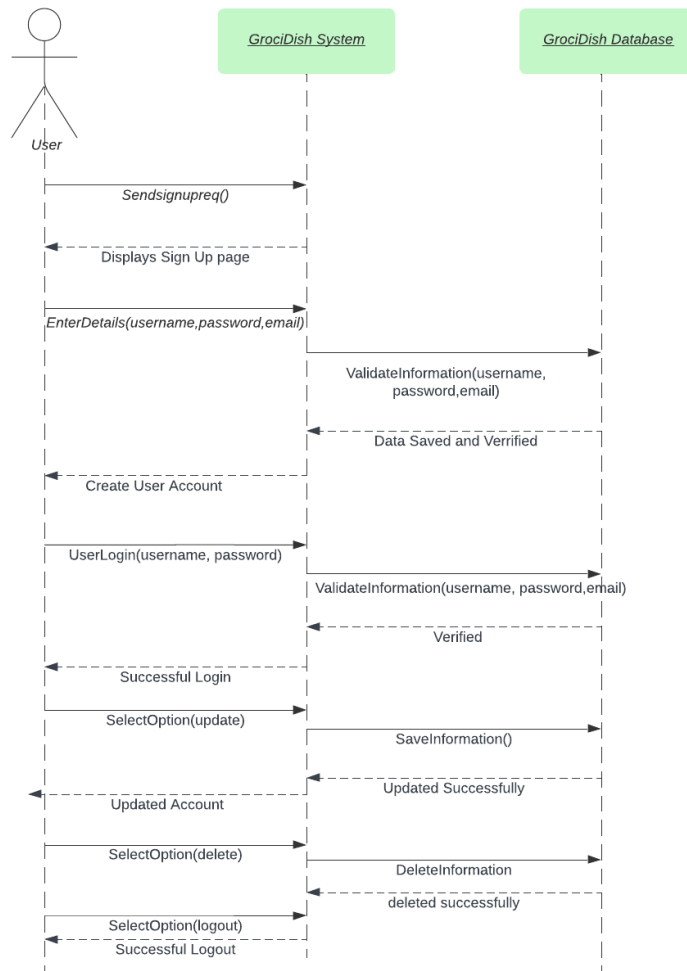


Figure 2.2: Manage Account SSD

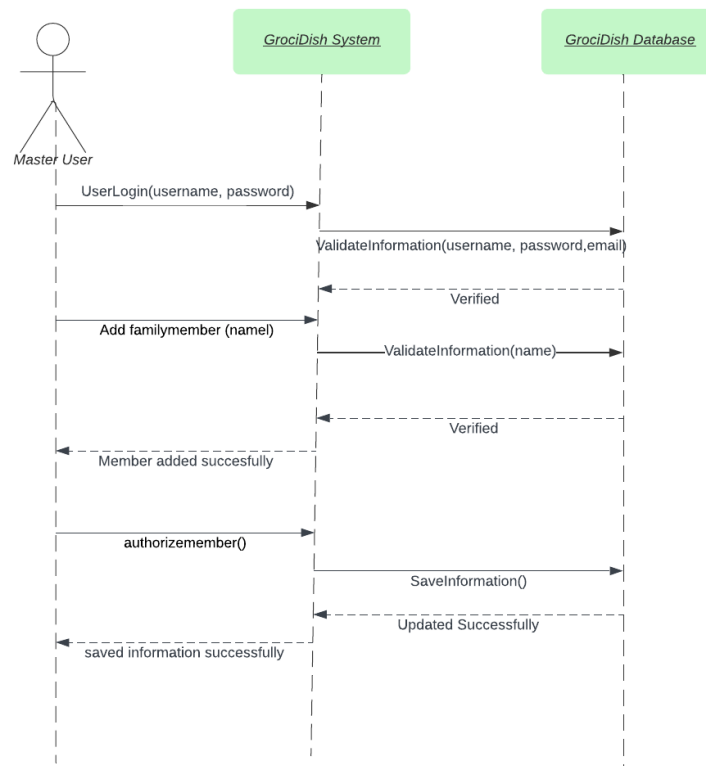


Figure 2.3: Manage Family Members SSD

2.3.0.2 Manage Profile

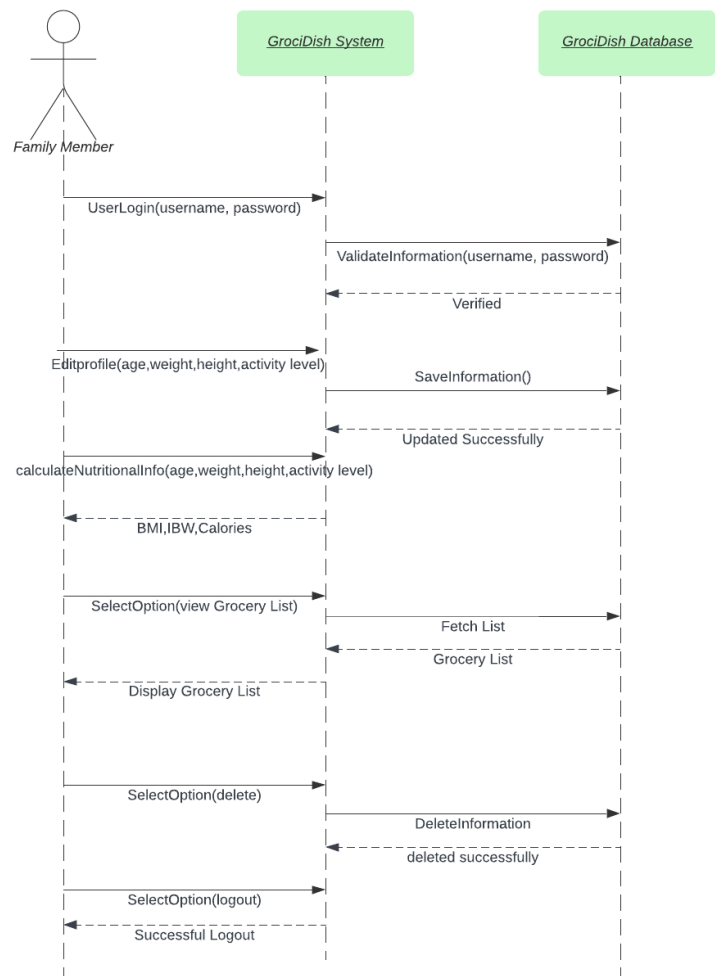


Figure 2.4: Manage Profile SSD

2.3.0.3 Track Calories

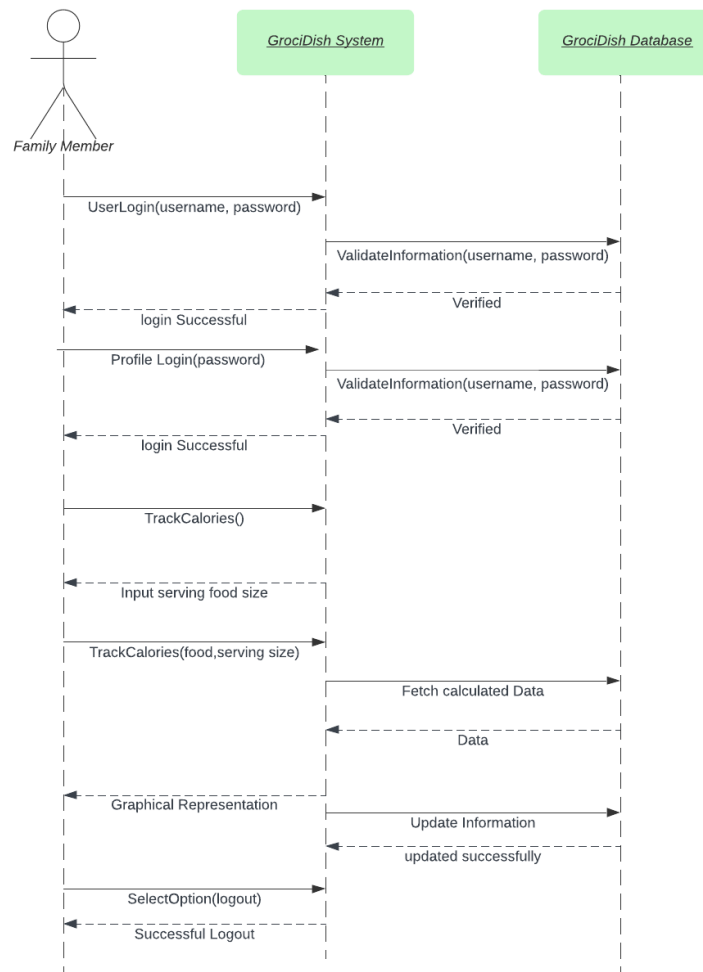


Figure 2.5: Track Calories SSD

2.3.0.4 Generate and Modify Grocery List

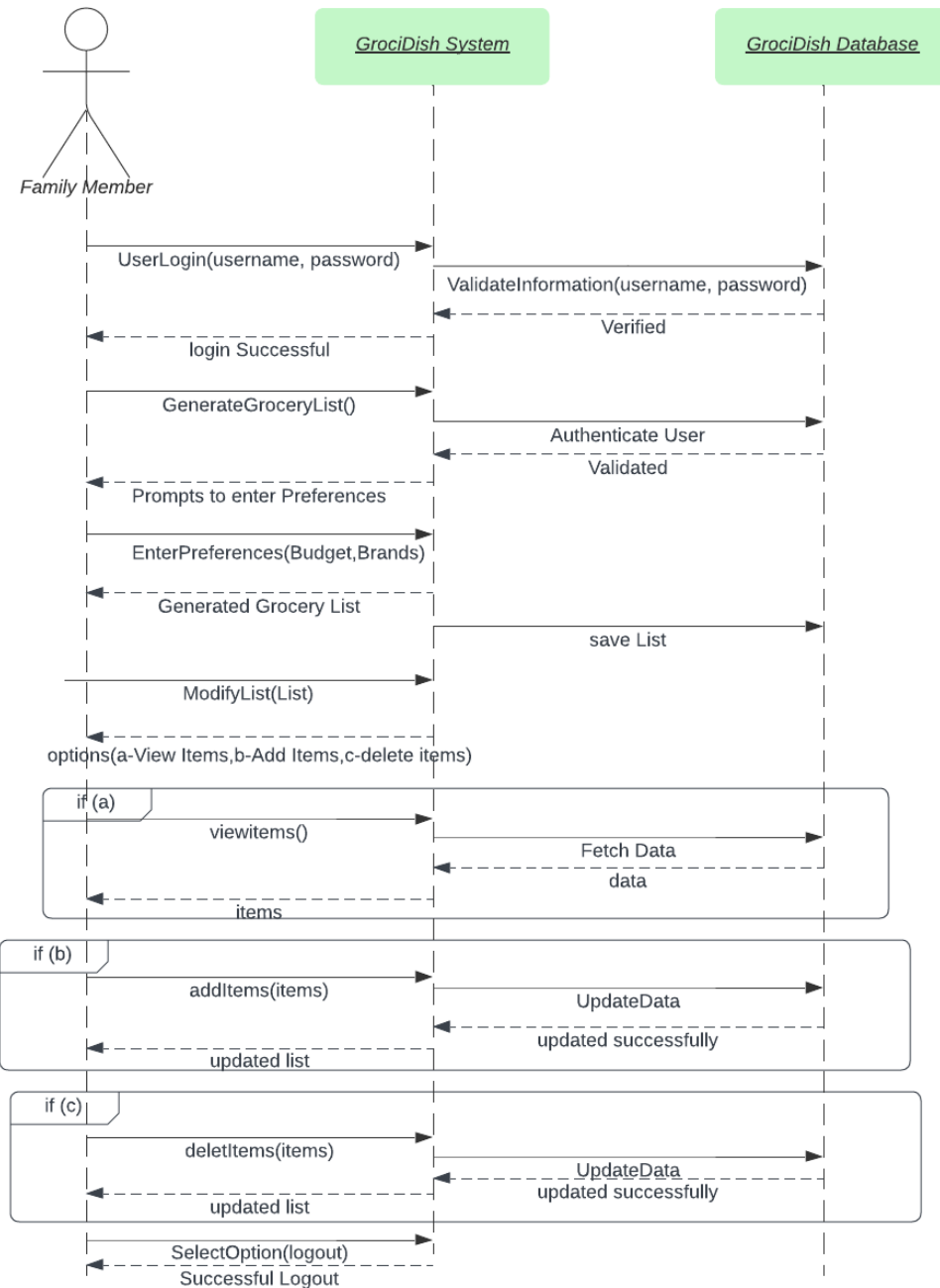


Figure 2.6: Generate and Modify Grocery List SSD

2.3.0.5 Select Meal Plan

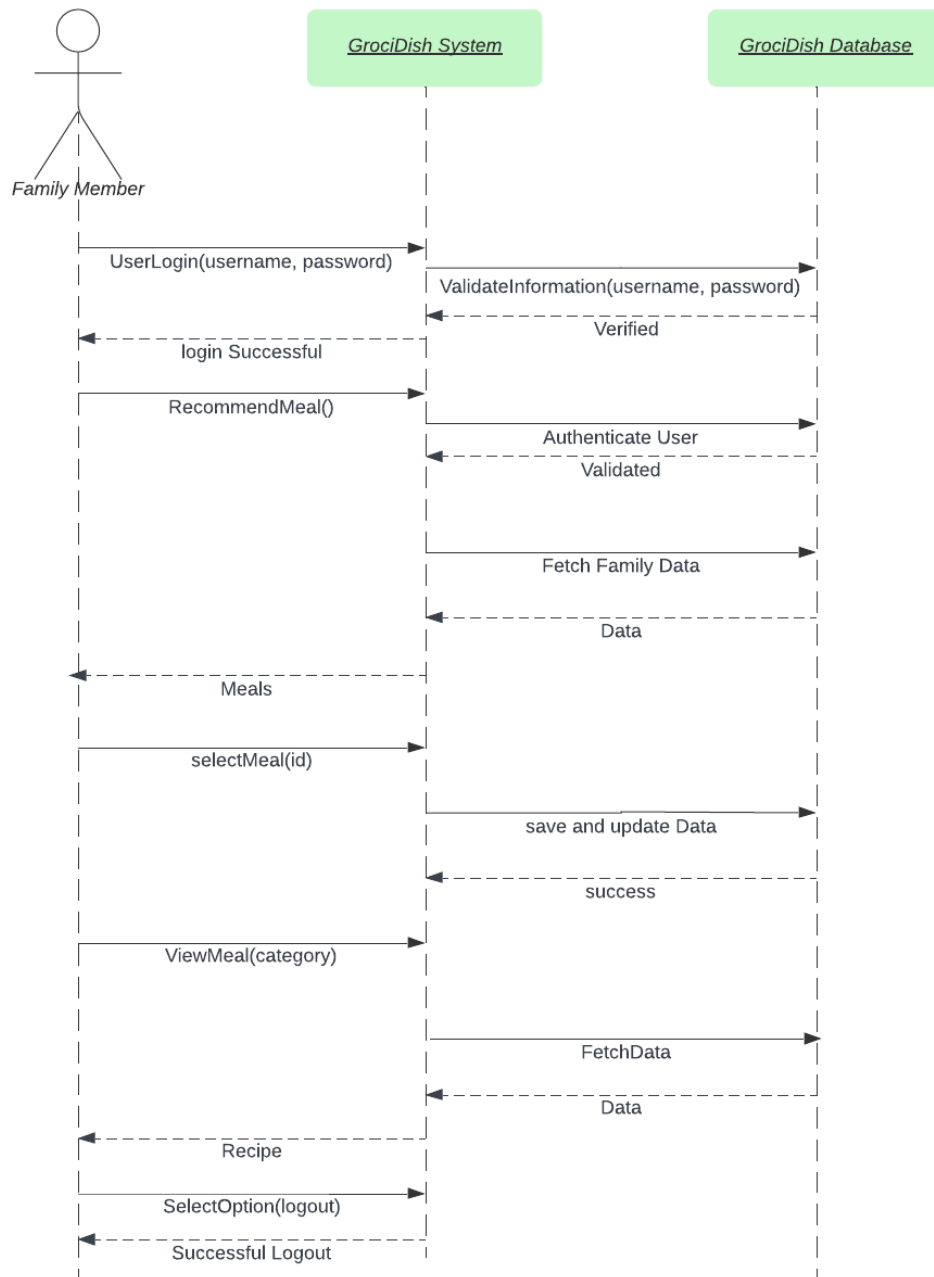


Figure 2.7: Select Meal Plan SSD

2.3.0.6 View Recipe Catalog

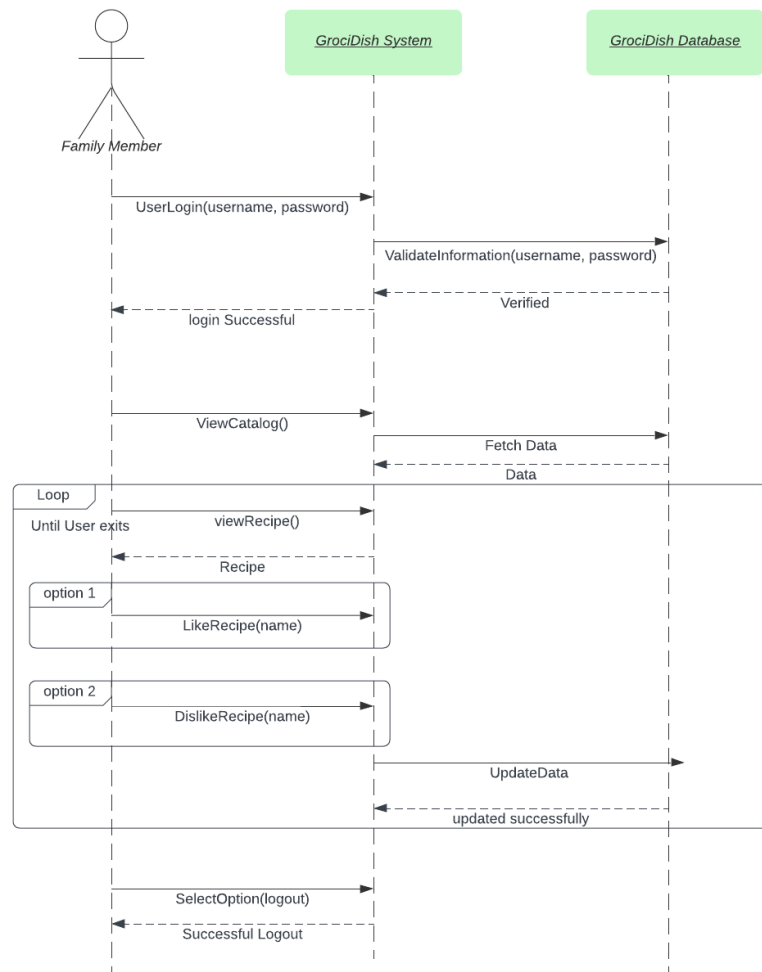


Figure 2.8: View Recipe Catalog SSD

2.4 Functional Requirements

This section describes the functional requirements of the GrociDish App.

2.4.1 Module 1: Manage Account

Following are the requirements for Module 1:

1. FR1: The system shall allow users to register a new account with valid email, user-name, and password.
2. FR2: The system shall validate user registration data and securely store it in the database.

3. FR3: The system shall allow users to add family members and assign permissions for grocery list generation and recipe selection.
4. FR4: The system shall confirm account creation and family member addition upon successful registration.
5. FR5: The system shall allow users to log in using their registered credentials.
6. FR6: The system shall allow users to update and delete their account after successful login.

2.4.2 Module 2: Manage Profile

Following are the requirements for Module 2:

1. FR1: The system shall allow users(Family Members) to view their profile details, including name, age, and role etc.
2. FR2: The system shall allow users to edit their profile information.
3. FR3: The system shall allow users to change their password for security purposes.
4. FR4: The system shall allow the Master User to update family member details and manage their roles.

2.4.3 Module 3: Personalized Grocery List Generation

Following are the requirements for Module 3:

1. FR1: The system shall allow authorized user to create personalized grocery lists based on their preferences.
2. FR2: The system shall incorporate budget constraints to generate cost-effective grocery lists.
3. FR3: The system shall provide options for brand preferences, allowing users to select grocery brands.

2.4.4 Module 4: Grocery List Modification

Following are the requirements for Module 4:

1. FR1: The system shall allow users to append items(add,remove) to their grocery list via a chat interface.
2. FR2: The system shall retrieve the current grocery list, providing users with an updated view of the items added.
3. FR3: The system shall incorporate natural language processing to enhance user interaction with the chatbot.

2.4.5 Module 5: Meal Recommendation

Following are the requirements for Module 5:

1. FR1: The system shall suggest recipes based on ingredients from the user's grocery list.
2. FR2: The system shall allow users to select Meal suggestions by health conditions like diabetes, hypertension, and obesity.
3. FR3: The system shall include alternative Meal options for variety while maintaining nutritional balance.
4. FR4: The system shall perform nutritional calculations (total calories) for accuracy and suitability.
5. FR5: The system shall provide access to a comprehensive database of pre-existing recipes.

2.4.6 Module 6: Nutritional Recommendation

Following are the requirements for Module 6:

1. FR1: The system shall take user input including age, weight, height, gender, and activity level.
2. FR2: The system shall calculate BMI (Body Mass Index) and IBW (Ideal Body Weight) for health assessments.
3. FR3: The system shall use the Harris-Benedict equation to calculate daily calorie needs based on activity levels.
4. FR4: The system shall recommend for each family member, detailing daily caloric, carbohydrate, protein, and fat intake.

2.4.7 Module 7: Daily Caloric Tracking

Following are the requirements for Module 7:

1. FR1: The system shall allow each family member to have a password-protected calorie tracker for privacy.
2. FR2: The system shall track daily caloric intake, providing users with a comprehensive view of their diet.
3. FR3: The system shall provide a visual breakdown of calorie sources, identifying unhealthy eating patterns or nutrient deficiencies.

2.5 Non-Functional Requirements

This section specifies nonfunctional requirements of the GrociDish App.

2.5.1 Reliability

Error Handling: The system should manage expected and unexpected errors efficiently, providing users with clear error messages, such as when a grocery list or recipe fails to generate.

Fault Tolerance: GrociDish should handle failures like a service or database crash by using retries and backups for key processes, including grocery list generation and recipe recommendations.

2.5.2 Usability

Ease of Use: The app interface (React Native) must be intuitive and user-friendly, allowing easy navigation for functions like updating grocery lists and viewing recipes. .

Interoperability: Usability requirements ensure ease of use, error prevention, efficient interaction, and accessibility for all users. These requirements help design an optimal user experience.

2.5.3 Performance

Data Processing: GrociDish must handle real-time calorie tracking and ingredient updates as meals are consumed by family members.

Background Tasks: Weekly scraping of grocery prices must not affect real-time performance and should be processed asynchronously.

- **PER-1:** The system shall generate a personalized grocery list within **10 seconds** after user input.
- **PER-2:** Updates to the grocery list shall be reflected in real-time, with changes processed in **less than 10 seconds**.
- **PER-3:** The chatbot shall respond to user queries within **5 seconds**.
- **PER-4:** The system shall suggest Meal plan based on ingredients within **10 seconds**.
- **PER-5:** Calorie recommendations shall be processed within **5 seconds**.
- **PER-6:** BMI and IBW calculations shall achieve **95% accuracy** based on user data.
- **PER-7:** Calorie breakdowns shall be displayed within **5 seconds** after user request.
- **PER-8:** Login requests shall be processed within **3 seconds**.

2.5.4 Security

Authorization and Role Management: Only authorized users should generate grocery lists or select Meal Plans. Each family member should have appropriate role-based access.

Authentication: The system should implement secure login with JWT, ensuring token expiration and revocation mechanisms. Security requirements should protect the system and its data from unauthorized access, defining the effort required to breach the system.

2.6 Domain Model

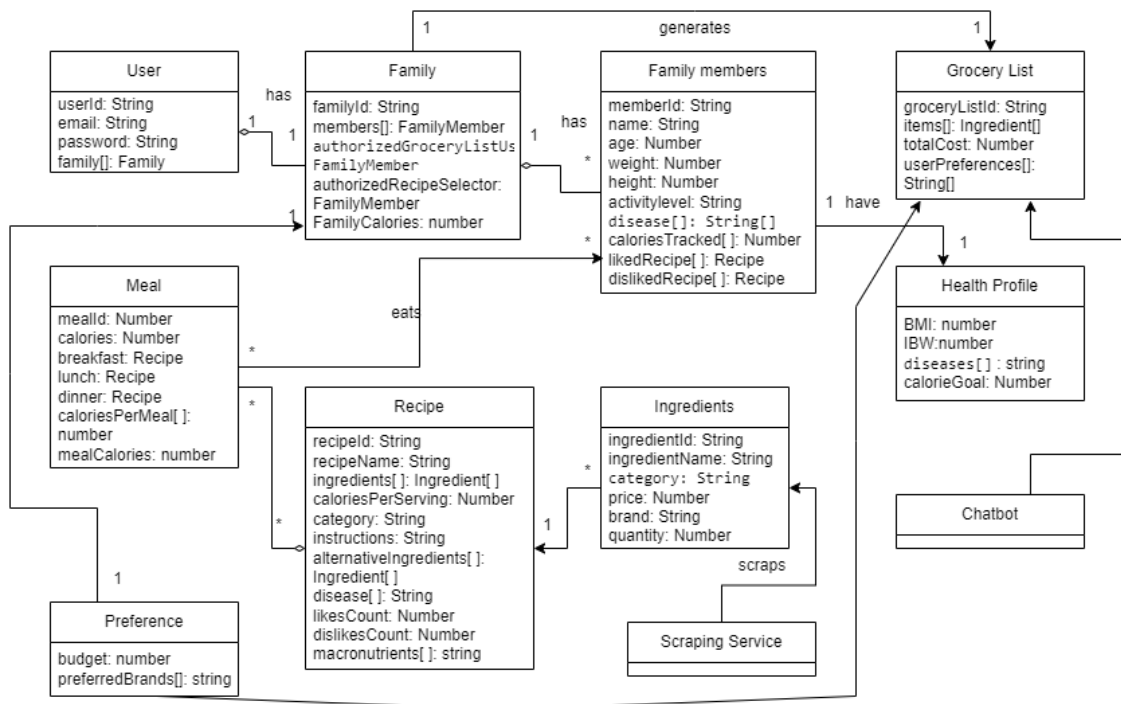


Figure 2.9: GrociDish Domain Model

Chapter 3

System Overview

3.1 Architectural Design

The architecture diagram for **GrociDish** shows the interaction between the **frontend application**, backend services, and the **AI models**. Users access the system through the **frontend application**, which communicates with the **API Gateway**. The gateway acts as the central hub for routing requests to the various backend services, including **User & Family Management**, **Grocery List Service**, **Meal Tracking Service**, **Recipe Management**, and the **Chatbot Service**. These services interact with a **database** to manage users, recipes, and grocery data. The **AI services** — specifically the **Grocery Generation Model** and the **Recipe Suggestion Model** — provide tailored recommendations for groceries and recipes, enhancing the app's functionality based on user data. The architecture ensures a modular design with AI integration for personalized recommendations.

3.1.1 Architectural Diagram

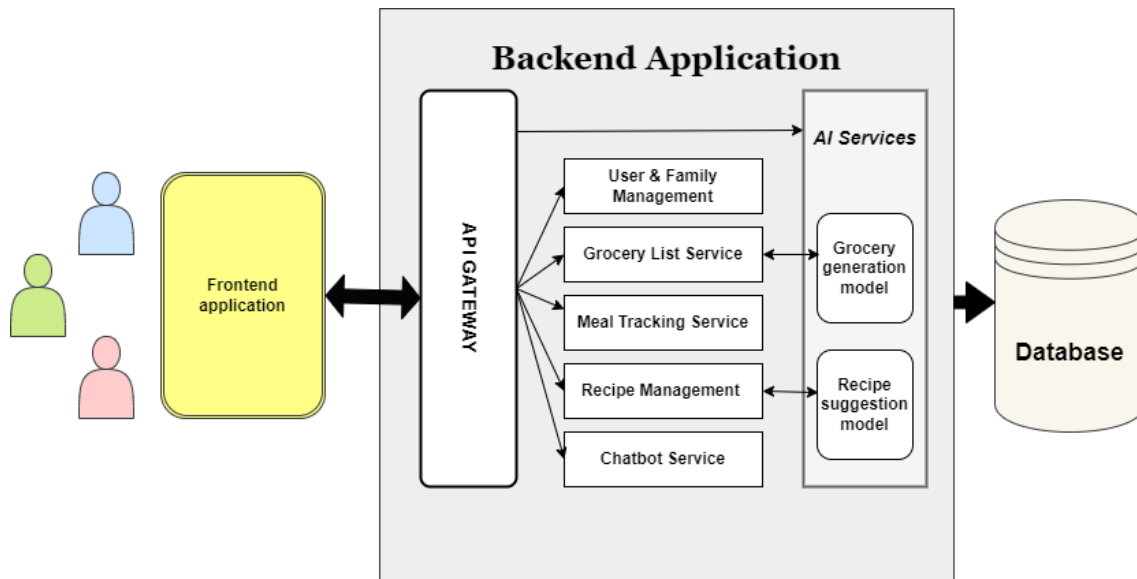


Figure 3.1: GrociDish Architectural Diagram

3.2 Design Models

Approach: Object Oriented Development Approach

OOP is better suited for GrociDish because it organizes the system into reusable, modular components that represent real-world entities like users, recipes, and grocery lists. This structure enhances maintainability and scalability as the app grows, allowing features like calorie tracking, recipe selection, and family member management to be handled efficiently within separate classes. OOP also enables encapsulation, securing sensitive data like user health information, and supports re-usability through inheritance, making it easier to extend functionality in the future without major rewrites. This flexibility makes OOP ideal for a complex, evolving system like GrociDish.

3.2.1 Activity Diagram

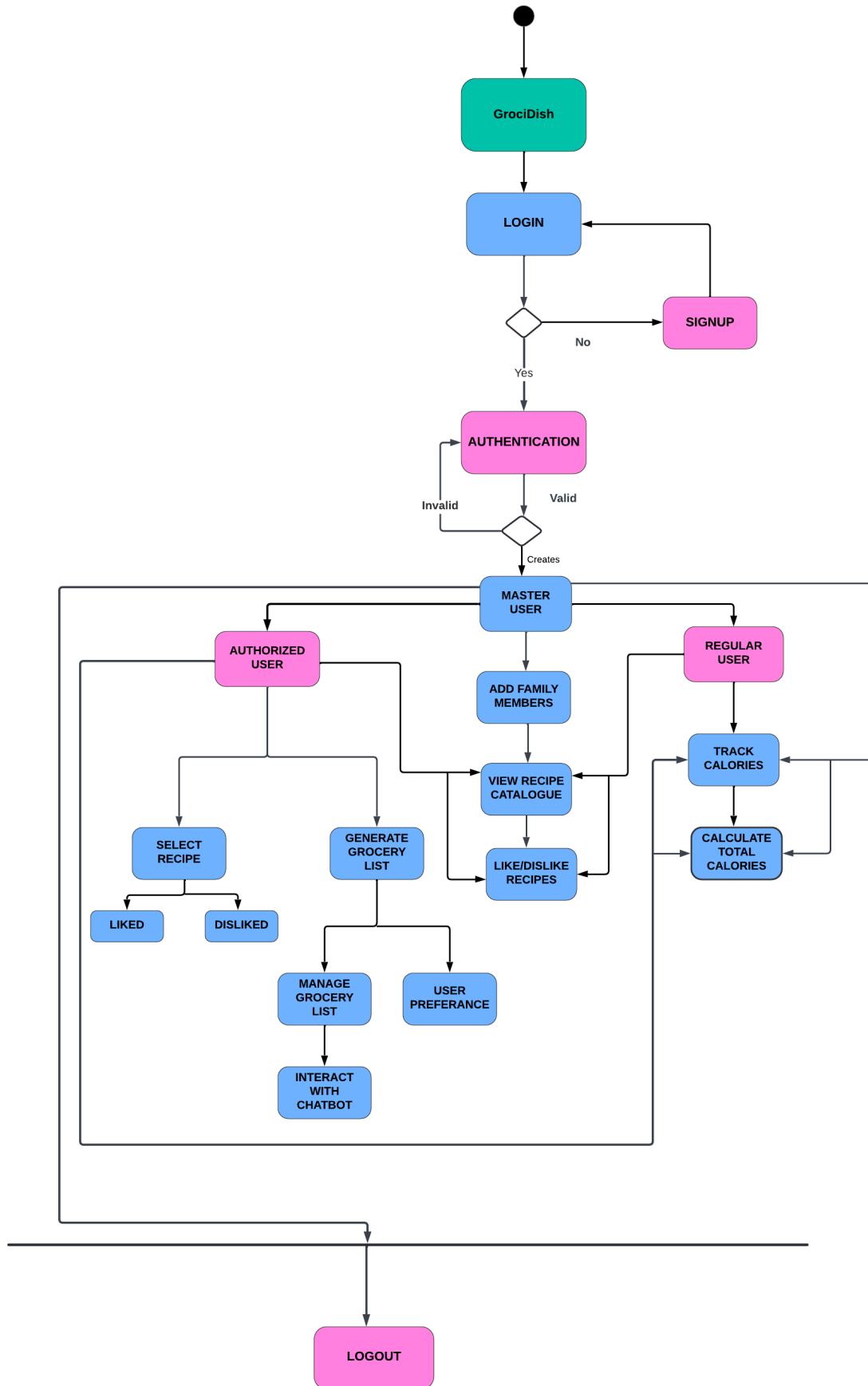


Figure 3.2: GrociDish Activity Diagram

3.2.2 Class Diagram

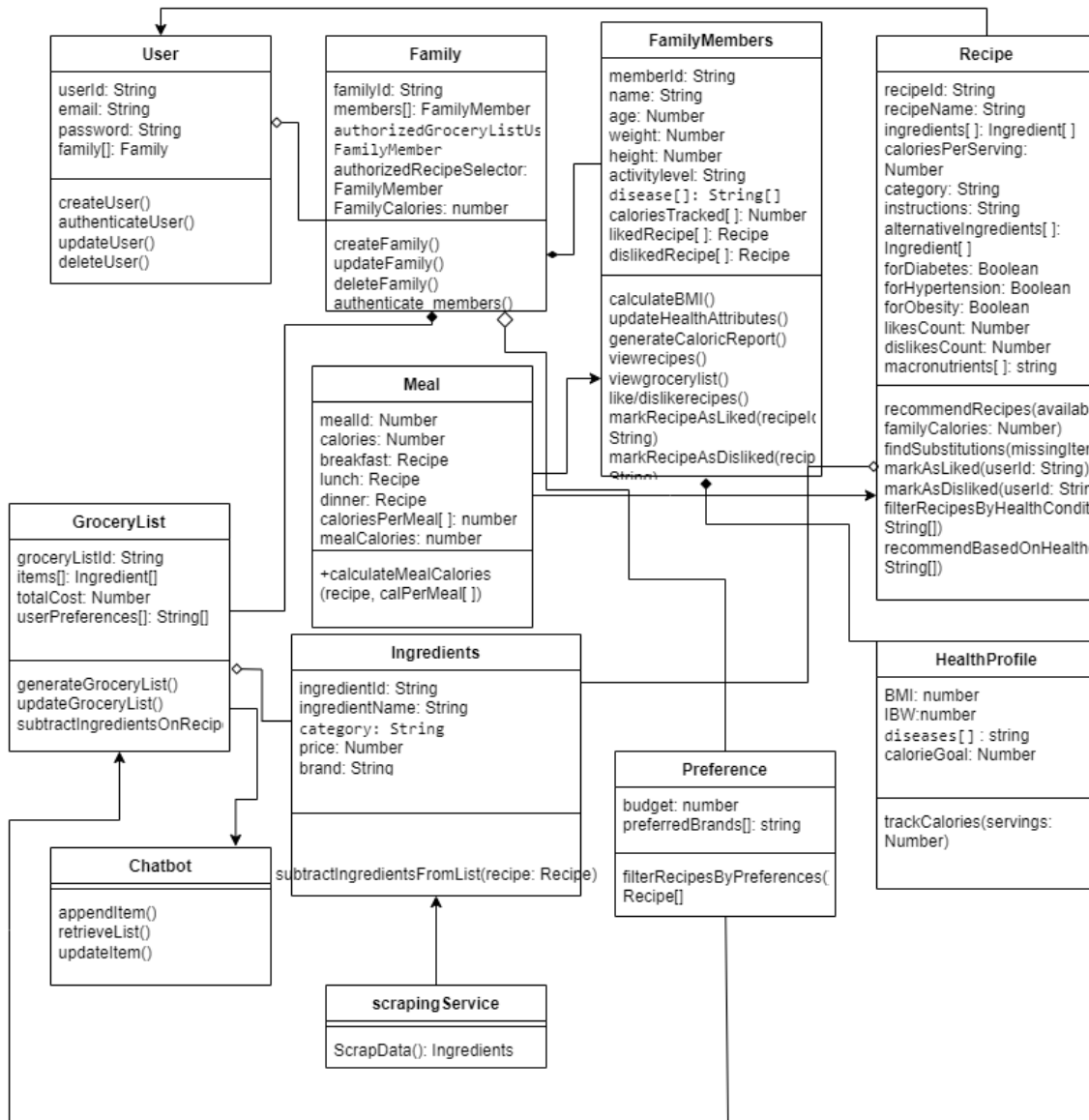


Figure 3.3: GrociDish Class Diagram

3.2.3 Class Level Sequence Diagrams

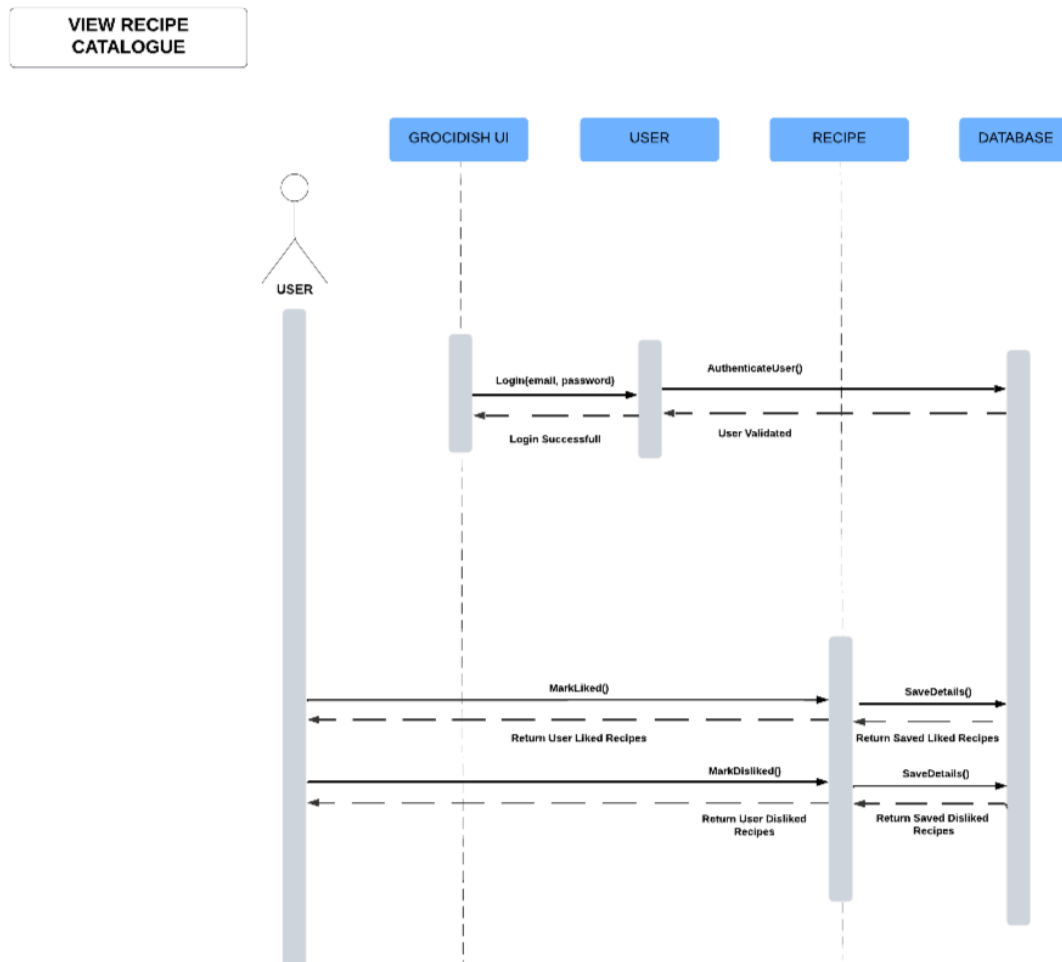


Figure 3.4: View Recipe Catalog SD

REGISTER ACCOUNT

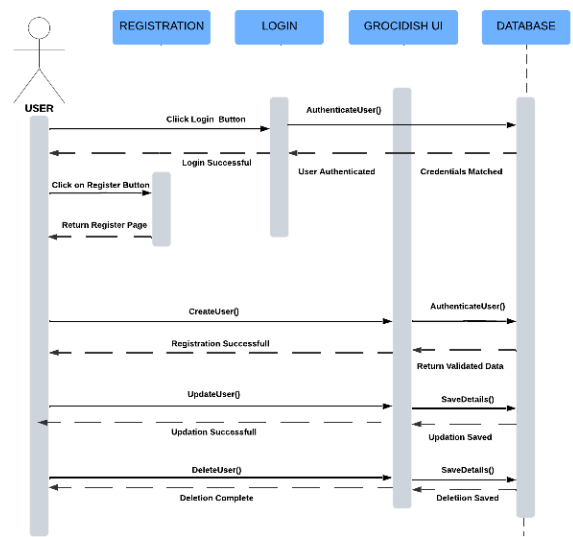


Figure 3.5: Register Account SD

GENERATE GROCERY LIST

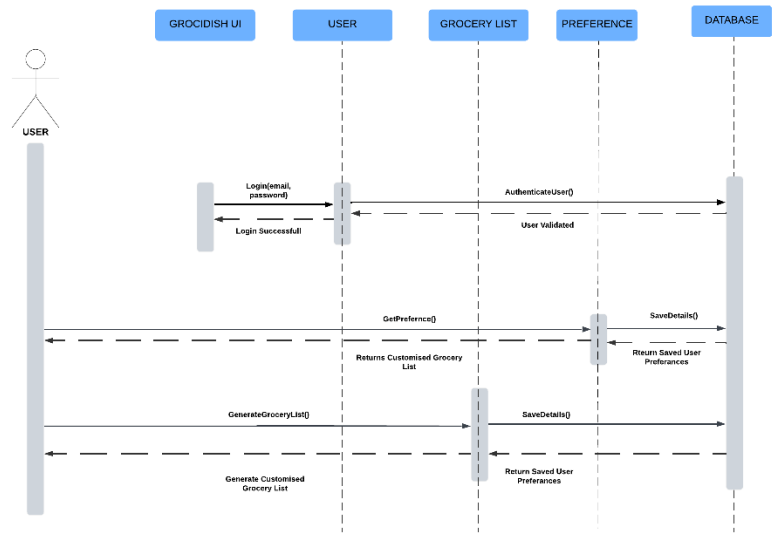


Figure 3.6: Generate Grocery List SD

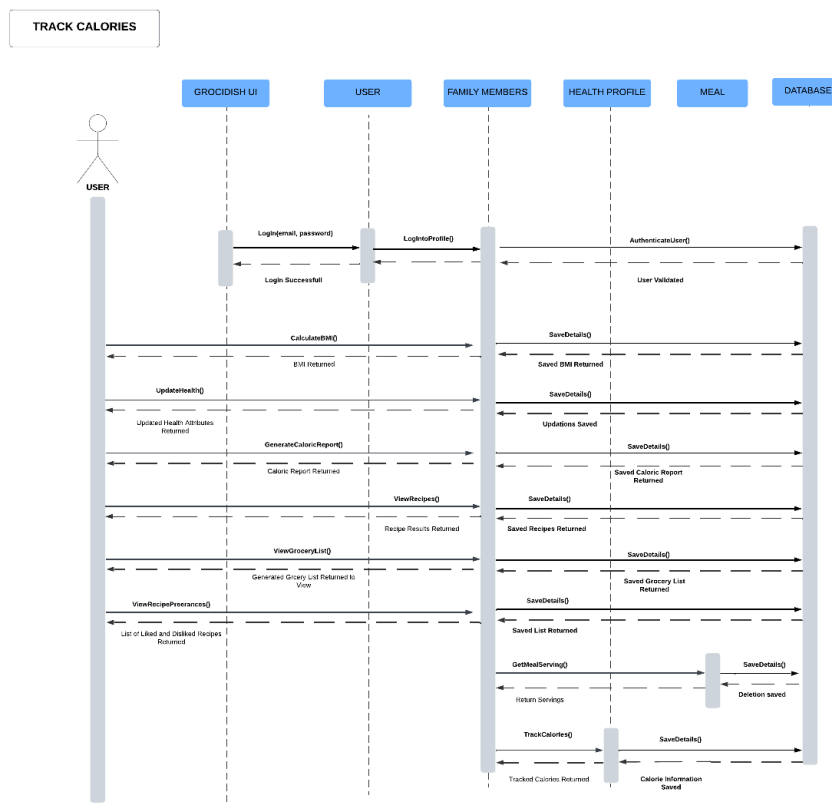


Figure 3.7: Track Calories SD

MODIFY GROCERY LIST

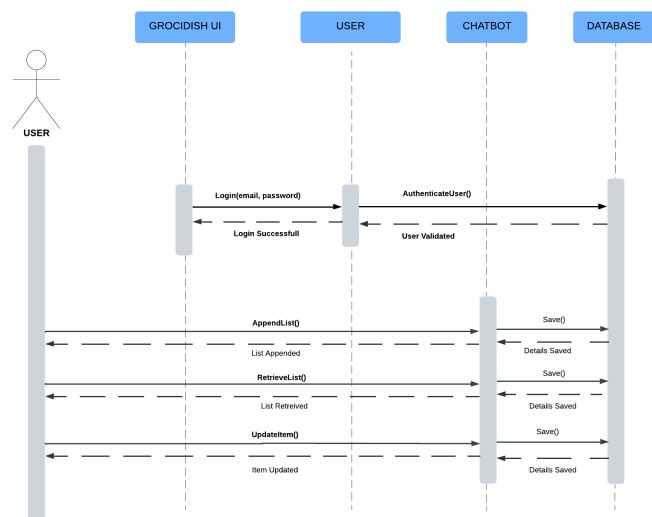


Figure 3.8: Select Meal SD

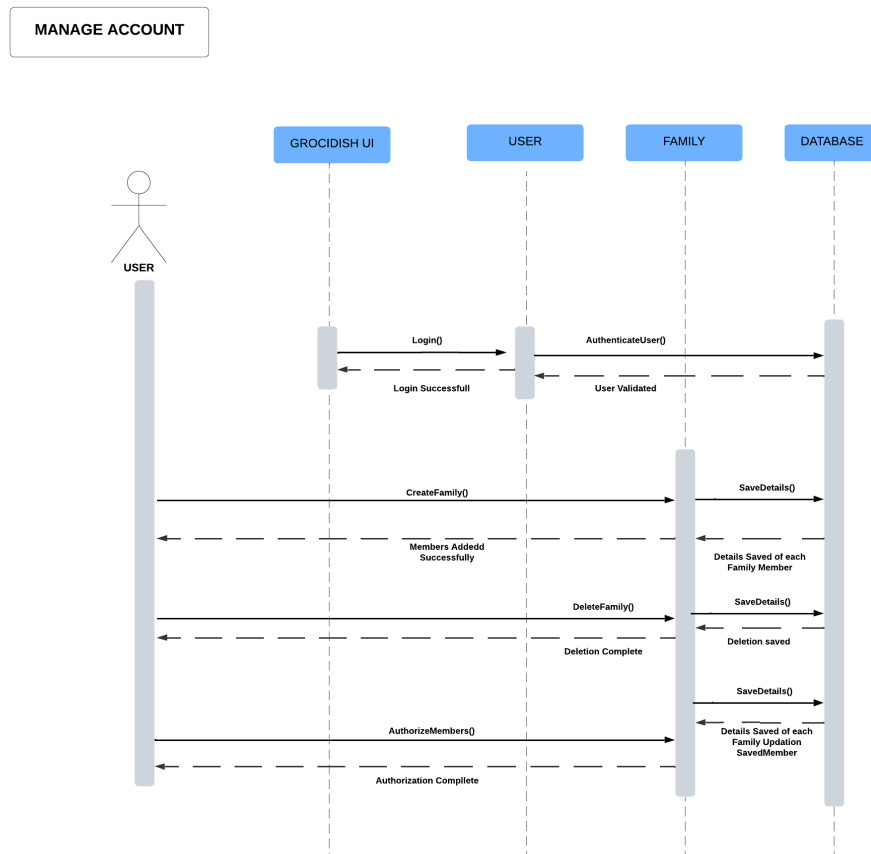


Figure 3.9: Manage Account SD

MODIFY GROCERY LIST

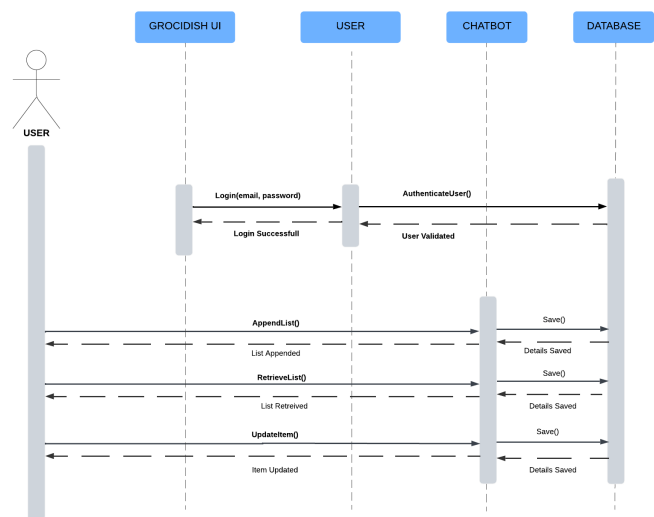


Figure 3.10: Modify Grocery List SD

3.3 Data Design

GrociDish utilizes PostgreSQL as its primary database management system to efficiently store and manage user profiles, grocery items, recipes, and health data. The system follows a relational model, ensuring robust data integrity and support for complex queries. Key entities like Users, Grocery-Items, Recipes, and Health-Profile are linked through well-defined relationships to facilitate seamless data retrieval and storage. The ERD diagram below illustrates the relationships between these core entities in the GrociDish database structure.

3.3.1 ERD Diagram

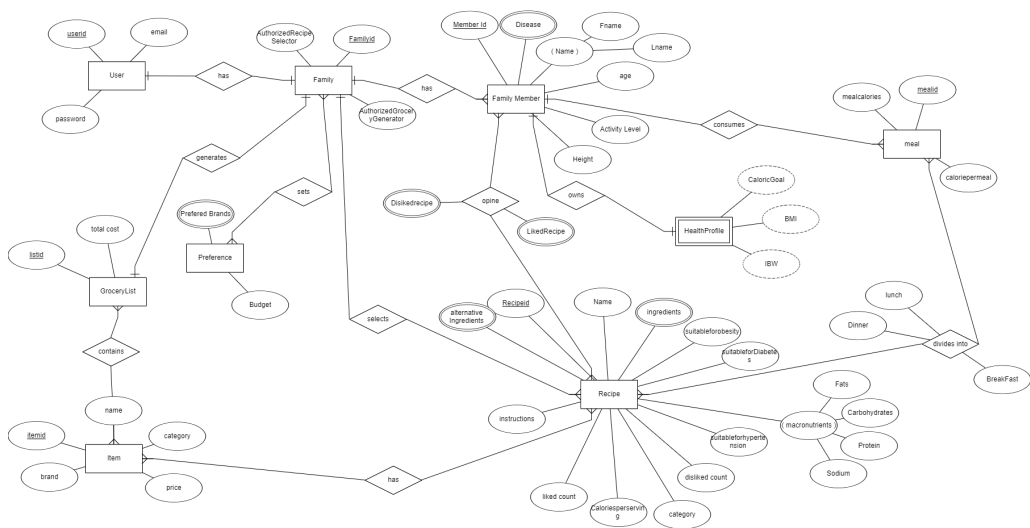


Figure 3.11: GrociDish ERD Diagram